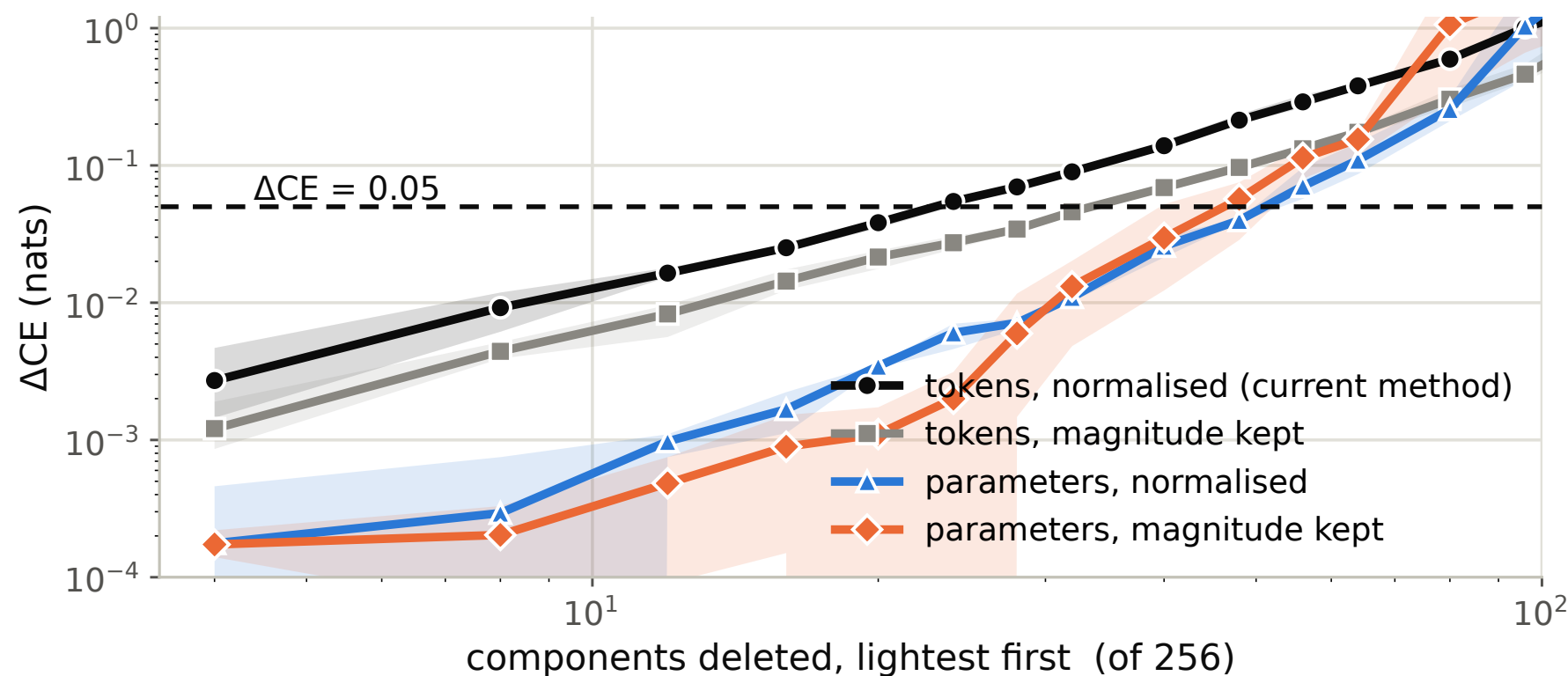


Clustering parameters by tokens, instead of tokens by parameters

The attribution matrix is $[N \text{ tokens} \times D \text{ weight coordinates}]$. The production fit clusters its ROWS — tokens that use similar parameters — and derives weight ownership from that. The transpose clusters its COLUMNS: parameters used by similar tokens. Crossed with whether magnitude survives normalisation, that gives four arms, each run through the full pipeline (GIM features → k-means → softpart bank → ablation) at $C=256$ on a 8k-position pilot, three seeds. The transposed k-means costs 0.1–0.5 s, cheaper than the row k-means it replaces; the bank build dominates at ~ 40 s.

How much can be deleted before behaviour moves



Transposing the clustering more than doubles it

